Lecture Scribe: Probability

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PDF

You are a course expert in computation probability and act as a scholar.
I am providing you lecture PowerPoint presentation. All you have to do is to convert it into a proper and a detailed version of it in lecture scribe format. The lecture scribe should contain:
Make sure to identify and cover the main concepts, definitions, formulae, diagrams and examples
Infer explanations from each topic and give a detailed note on it.
For each slide/topic, generate:
Topic heading and highlight the important topics
Explain the concepts in the simple language and easy words with proper examples
Include derivation steps
Explain logic in simple words
Include all the applications given in the ppt
Split down the complex concepts into step by step and simpler explanations
Do explain using conceptual approach and mathematic approach
If you find any inconsistency, solve it and generate the most accurate explanation
Give summary and make sure to include important topics
Give the formulas sheet
Do not assume prior knowledge
Do not skip any slide and topic

Awesome, this is a great lecture to turn into a proper **lecture scribe**. I'll rewrite the entire PPT into **clean, detailed, student-friendly notes**, covering **every slide**, explaining **concepts from zero**, with **derivations, intuition, examples, applications, summaries, and a formula sheet** — exactly like you asked.

All content below is based on your uploaded lecture slides on **Gaussian, Uniform, Exponential, Laplace, and Gamma Random Variables**

CSE400 – Fundamentals of Probability in Computing

Lecture 8: Gaussian, Uniform, Exponential, Laplace, and Gamma Random Variables (Complete Lecture Scribe)

1. Introduction and Outline of the Lecture

This lecture focuses on **important continuous random variables** that are widely used in **probability, statistics, signal processing, machine learning, and computing systems**.

Main topics covered:

- Gaussian (Normal) Random Variable
  - Definition and properties
  - Standard forms: Error function, Complementary error function
  - Φ-function (CDF) and Q-function (Tail probability)
  - Relationship between Φ and Q
  - Example problems
- Types of Continuous Random Variables
  - Uniform Random Variable
  - Exponential Random Variable
  - Laplace Random Variable
  - Gamma Random Variable (with graph and special cases)
- Problem solving and applications
- In-class activity: Gaussian density estimation

Before starting, we recall:

A **random variable (RV)** is a variable whose value depends on the outcome of a random experiment.

- If it takes **discrete values** → Discrete RV (uses PMF)
- If it takes **continuous values** → Continuous RV (uses PDF and CDF)

For a continuous random variable X:

- PDF: $f_X(x)$
- CDF: $F_X(x) = P(X \leqslant x) = \int_{-\infty}^{x} f_X(t)\, dt$

2. Expectation and Moments (Background from Slides)

2.1 Expected Value of a Function of a Random Variable

For a **continuous random variable**:

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$$

For a **discrete random variable**:

$$E[g(X)] = \sum_k g(x_k) P(X = x_k)$$

Special cases:

- Mean (first moment):

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx$$

- Variance:

$$\mathrm{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$$

where $\mu = E[X]$

2.2 Linearity of Expectation

For constants a, b:

$$E[aX + b] = aE[X] + b$$

For sum of random variables:

$$E[X_1 + X_2 + \cdots + X_N] = E[X_1] + E[X_2] + \cdots + E[X_N]$$

This property is **very important** because it works **even if variables are dependent**.

2.3 Higher Order Moments: Skewness and Kurtosis

- **Skewness** (3rd standardized moment):

$$C_3 = \frac{E[(X - \mu)^3]}{\sigma^3}$$

It measures **asymmetry** of the PDF:

- Positive → tail on right side
- Negative → tail on left side
- **Kurtosis** (4th standardized moment):

$$C_4 = \frac{E[(X-\mu)^4]}{\sigma^4}$$

It measures **peakedness**:

- Large value → sharper peak near mean
- Smaller value → flatter distribution

---

3. Gaussian (Normal) Random Variable

3.1 Definition

A random variable X is called **Gaussian (Normal)** if its PDF is:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Where:

- μ = mean (center of the distribution)
- σ^2 = variance
- σ = standard deviation

We write:

$$X \sim \mathrm{N}(\mu, \sigma^2)$$

3.2 Interpretation of Parameters

- μ: shifts the curve left or right
- σ: controls the **spread**
  - Small σ → narrow, tall curve
  - Large σ → wide, flat curve

3.3 Shape and Properties

- Bell-shaped curve
- Symmetric about μ
- Total area under the PDF = 1

- Mean = Median = Mode = μ

The CDF has an **S-shape**:

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$$

---

4. Standard Normal Distribution

If:

$$Z = \frac{X - \mu}{\sigma}$$

Then:

$$Z \sim \mathrm{N}(0, 1)$$

This is called the **standard normal random variable**.

Its PDF:

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

This transformation is called **standardization** and is used to convert any Gaussian into standard form.

---

5. Error Function and Complementary Error Function

5.1 Error Function (erf)

Defined as:

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_{0}^{x} e^{-t^2}\,dt$$

It appears when integrating Gaussian-type expressions.

5.2 Complementary Error Function (erfc)

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_{x}^{\infty} e^{-t^2}\,dt$$

These functions are commonly used in:

- Communication systems
- Signal detection
- Probability of error analysis

---

6. Φ-function and Q-function

6.1 Φ-function (CDF of Standard Normal)

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives:

$$\Phi(x) = P(Z \leqslant x)$$

It represents **area under the curve to the left of x**.

6.2 Q-function (Gaussian Tail Function)

$$Q(x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives:

$$Q(x) = P(Z > x)$$

It represents **area under the curve to the right of x**.

6.3 Relationship between Φ and Q

Since total area = 1:

$$Q(x) = 1 - \Phi(x)$$

---

7. CDF of a General Gaussian Random Variable

Let:

$$X \sim \mathrm{N}(\mu, \sigma^2)$$

Then:

$$F_X(x) = P(X \leqslant x)$$

Derivation:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-\mu)^2/(2\sigma^2)} dy$$

Substitute:

$$t = \frac{y-\mu}{\sigma}$$

Then:

$$F_X(x) = \int_{-\infty}^{(x-\mu)/\sigma} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

So:

$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

7.1 Tail Probability

$$P(X > x) = Q\left(\frac{x-\mu}{\sigma}\right)$$

Key identities:

$$Q(x) = 1 - \Phi(x)$$

$$F_X(x) = 1 - Q\left(\frac{x-\mu}{\sigma}\right)$$

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8. Worked Example (From Slides)

Given:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Compare with:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$$

We identify:

$$\mu = -3, \quad \sigma^2 = 4, \quad \sigma = 2$$

So:

$$X \sim \mathrm{N}(-3, 4)$$

(1) Find $P(X \leqslant 0)$

$$P(X \leqslant 0) = F_X(0) = \Phi\left(\frac{0-(-3)}{2}\right) = \Phi\left(\frac{3}{2}\right)$$

Or:

$$= 1 - Q\left(\frac{3}{2}\right)$$

---

(2) Find $P(X > 4)$

$$P(X > 4) = Q\left(\frac{4-(-3)}{2}\right) = Q\left(\frac{7}{2}\right)$$

---

(3) Probabilities of intervals

For any:

$$P(a < X < b) = F_X(b) - F_X(a) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

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9. Other Important Continuous Random Variables

9.1 Uniform Random Variable

Definition:

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leqslant x \leqslant b \\ 0, & \text{otherwise} \end{cases}$$

Properties:

- All values in $[a, b]$ are **equally likely**
- Mean:

$$E[X] = \frac{a+b}{2}$$

- Variance:

$$\mathrm{Var}(X) = \frac{(b-a)^2}{12}$$

Application:

- Random number generation
- Simulation
- Modeling equal uncertainty

9.2 Exponential Random Variable

Definition:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geqslant 0 \\ 0, & x < 0 \end{cases}$$

Properties:

- Mean:

$$E[X] = \frac{1}{\lambda}$$

- Variance:

$$\mathrm{Var}(X) = \frac{1}{\lambda^2}$$

- Memoryless property:

$$P(X > s + t \mid X > s) = P(X > t)$$

Applications:

- Waiting time for events
- Queueing systems
- Failure time modeling

9.3 Laplace Random Variable

PDF:

$$f_X(x) = \frac{1}{2b} e^{-|x-\mu|/b}$$

Properties:

- Sharp peak at μ
- Heavier tails than Gaussian
- Used in:
  - Noise modeling
  - Differential privacy
  - Signal processing

9.4 Gamma Random Variable

PDF:

$$f_X(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}, \quad x \geqslant 0$$

Parameters:

- Shape: k
- Scale: θ

Mean:

$$E[X] = k\theta$$

Variance:

$$\mathrm{Var}(X) = k\theta^2$$

Special cases:

- Exponential distribution (when $k = 1$)
- Erlang distribution (when k is integer)

Applications:

- Waiting time for multiple events
- Reliability analysis

- Queueing systems

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10. Summary of the Lecture

- Gaussian distribution is the **most important continuous distribution**
- Defined by mean μ and variance σ^2
- Standard normal uses Φ-function and Q-function
- Relationship:
  - $F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$
  - $P(X > x) = Q\left(\frac{x-\mu}{\sigma}\right)$
- Error function and complementary error function appear in Gaussian integrals
- Uniform, Exponential, Laplace, and Gamma are other important continuous RVs with real-world applications

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11. Formula Sheet (Quick Reference)

Gaussian PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$$

Standardization:

$$Z = \frac{X-\mu}{\sigma}$$

CDF:

$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

Q-function:

$$Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Relation:

$$Q(x) = 1 - \Phi(x)$$

Uniform:

$$f_X(x) = \frac{1}{b-a}, \ a \leqslant x \leqslant b$$

Exponential:

$$f_X(x) = \lambda e^{-\lambda x}, \ x \geqslant 0$$

Laplace:

$$f_X(x) = \frac{1}{2b} e^{-|x-\mu|/b}$$

Gamma:

$$f_X(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}$$

If you want, I can now:

- Convert this into **LaTeX notes**
- Turn it into a **proper exam-ready PDF**
- Or make a **short revision version** for quick study before tests.

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Perfect — I've carefully gone through your uploaded lecture slides and converted **every slide** into a **complete, detailed, student-friendly lecture scribe**, following **exactly** your instructions: definitions, intuition, derivations, diagrams (explained in words), examples, step-by-step logic, applications, summary, and a formula sheet. This is based on your file **Lecture 8: Gaussian, Uniform, Exponential, and Gamma Random Variables**

I will follow the slide order and **not skip anything**.

CSE400 – Fundamentals of Probability in Computing

Lecture 8: Gaussian, Uniform, Exponential, Laplace, and Gamma Random Variables (Lecture Scribe)

Slide 1: Title Slide

Topic: Gaussian, Uniform, Exponential, and Gamma Random Variables

This lecture introduces **important continuous random variables** that are used in:

- Probability and statistics
- Signal processing and communication systems
- Machine learning and data modeling

- Performance analysis in computing systems

Main focus:

- Gaussian (Normal) random variable and its special functions
- Other common continuous distributions: Uniform, Exponential, Laplace, Gamma
- How to compute probabilities using **CDF, Φ-function, and Q-function**
- Solved example using Gaussian distribution

Slide 2: Outline of the Lecture

The lecture is divided into:

1. Gaussian Random Variable
- Definition and properties
- Standard forms: Error function, Q-function
- Relation between Φ-function and Q-function
- Example
2. Types of Continuous Random Variables
- Uniform (with example)
- Exponential (with example)
- Laplace (with example)
- Gamma
  - Graph and special cases
  - Example
3. Problem solving
4. In-class activity: Gaussian density estimation

Before starting, we recall:

A **continuous random variable (CRV)** is a random variable that can take **any real value in an interval**.
It is described using:

- PDF (Probability Density Function)
- CDF (Cumulative Distribution Function)

Slide 3: Moments, Skewness, and Kurtosis

Skewness (3rd standardized moment)

$$C_3 = \frac{E[(X-\mu)^3]}{\sigma^3}$$

Meaning:

- Measures **asymmetry** of the PDF around the mean μ
- If skewness > 0 → tail is longer on the **right**
- If skewness < 0 → tail is longer on the **left**
- If skewness = 0 → distribution is **symmetric**

So skewness tells us **which side the distribution is leaning towards**.

Kurtosis (4th standardized moment)

$$C_4 = \frac{E[(X-\mu)^4]}{\sigma^4}$$

Meaning:

- Measures **peakedness** (how sharp or flat the peak is)
- Large kurtosis → very sharp peak near the mean
- Small kurtosis → flatter distribution

So kurtosis tells us **how concentrated the values are around the mean**.

Slide 4: Expected Value of a Function of a Random Variable

For a Continuous Random Variable (CRV)

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$$

This means:

- Multiply the function $g(x)$ with the PDF
- Integrate over all possible values
- This gives the **average value** of $g(X)$

For a Discrete Random Variable (DRV)

$$E[g(X)] = \sum_k g(x_k)P(X = x_k)$$

Linearity of Expectation

For constants a, b:

$$E[aX + b] = aE[X] + b$$

For sum of variables:

$$E\left[\sum_{k=1}^{N} g_k(X)\right] = \sum_{k=1}^{N} E[g_k(X)]$$

Important logic:

- Expectation of a sum = sum of expectations
- This is true **even if variables are dependent**

Slide 5: Gaussian Random Variable – Section Start

Now we start the **most important continuous distribution**: the Gaussian (Normal) distribution.

Slide 6–7: Definition of Gaussian Random Variable

A random variable X is **Gaussian** if its PDF is:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

We write:

$$X \sim \mathrm{N}(\mu, \sigma^2)$$

Where:

- μ = mean (center of the distribution)
- σ^2 = variance (spread)
- σ = standard deviation

Conceptual meaning:

- The curve is **bell-shaped**
- It is **symmetric** around μ
- Most values are near μ
- Fewer values appear as we move away from μ

The slide shows:

- PDF: bell-shaped curve
- CDF: S-shaped curve (area accumulated from left)

Applications (mentioned in slide annotations):

- Noise modeling (thermal noise)
- Measurement errors
- Natural variations (heights, errors, etc.)

Slide 8–10: Standard Forms – Error Function, Φ-function, Q-function

Error Function (erf)

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Used because Gaussian integrals **cannot be written in simple closed form**.

Complementary Error Function (erfc)

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt$$

Φ-function (CDF of Standard Normal)

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Meaning:

- $\Phi(x) = P(Z \leqslant x)$
- It gives **area under the curve to the left of x**

Q-function (Gaussian Tail Function)

$$Q(x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Meaning:

- $Q(x) = P(Z > x)$
- It gives **area under the curve to the right of x**

Important Relation

$$Q(x) = 1 - \Phi(x)$$

Because total area under the curve = 1.

Slide 11–17: Relation between Φ-function and Q-function (Derivation)

Let:

$$X \sim \mathrm{N}(\mu, \sigma^2)$$

CDF is:

$$F_X(x) = P(X \leqslant x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-\mu)^2/(2\sigma^2)} dy$$

Step 1: Standardization

Let:

$$t = \frac{y - \mu}{\sigma}$$

Then limits change:

- When $y = -\infty$, $t = -\infty$
- When $y = x$, $t = \frac{x-\mu}{\sigma}$

So:

$$F_X(x) = \int_{-\infty}^{(x-\mu)/\sigma} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Step 2: Recognize Φ-function

$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

Tail Probability

$$P(X > x) = \int_{(x-\mu)/\sigma}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt = Q\left(\frac{x-\mu}{\sigma}\right)$$

Key Identities (from slides)

$$Q(x) = 1 - \Phi(x)$$

$$F_X(x) = 1 - Q\left(\frac{x-\mu}{\sigma}\right)$$

Conceptual Meaning (from diagram slide)

- Φ(x) = area under **left tail** up to x
- Q(x) = area under **right tail** beyond x
- CDF questions → use Φ
- Right-tail probability questions → Q is more natural

Slide 18–20: Worked Example

Given PDF:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Compare with standard Gaussian:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

We identify:

$$\mu = -3, \quad \sigma^2 = 4, \quad \sigma = 2$$

So:

$$X \sim \mathrm{N}(-3, 4)$$

---

1) Find $P(X \leqslant 0)$

$$P(X \leqslant 0) = F_X(0) = \Phi\left(\frac{0-(-3)}{2}\right) = \Phi\left(\frac{3}{2}\right)$$

Using relation:

$$= 1 - Q\left(\frac{3}{2}\right)$$

---

2) Find $P(X > 4)$

$$P(X > 4) = Q\left(\frac{4-(-3)}{2}\right) = Q\left(\frac{7}{2}\right)$$

General Interval Probability

$$P(a < X < b) = F_X(b) - F_X(a)$$

Slides on Other Continuous Random Variables (From Outline)

Uniform Random Variable

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leqslant x \leqslant b \\ 0, & \text{otherwise} \end{cases}$$

Mean:

$$E[X] = \frac{a+b}{2}$$

Variance:

$$\mathrm{Var}(X) = \frac{(b-a)^2}{12}$$

Application:

- Random number generation
- Simulation
- Modeling equal uncertainty

Exponential Random Variable

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geqslant 0 \\ 0, & x < 0 \end{cases}$$

Mean:

$$E[X] = \frac{1}{\lambda}$$

Variance:

$$\mathrm{Var}(X) = \frac{1}{\lambda^2}$$

Special property: Memoryless

Application:

- Waiting time
- Queue systems
- Failure modeling

Laplace Random Variable

$$f_X(x) = \frac{1}{2b} e^{-|x-\mu|/b}$$

- Sharper peak than Gaussian
- Heavier tails
- Used in noise modeling and privacy mechanisms

Gamma Random Variable

$$f_X(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}, \quad x \geqslant 0$$

Mean:

$$E[X] = k\theta$$

Variance:

$$\mathrm{Var}(X) = k\theta^2$$

Special cases:

- Exponential (k = 1)
- Erlang (k integer)

Application:

- Waiting time for multiple events
- Reliability and queueing models

Final Summary

- Gaussian distribution is the **most important continuous distribution**
- Defined by mean μ and variance σ^2
- CDF uses Φ-function
- Tail probabilities use Q-function
- Relation: $Q(x) = 1 - \Phi(x)$
- Other important distributions: Uniform, Exponential, Laplace, Gamma
- These are widely used in computing, networks, and data modeling

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Formula Sheet

Gaussian PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$$

Standardization:

$$Z = \frac{X - \mu}{\sigma}$$

CDF:

$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

Q-function:

$$Q(x) = \int_x^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Relation:

$$Q(x) = 1 - \Phi(x)$$

Uniform:

$$f_X(x) = \frac{1}{b-a}$$

Exponential:

$$f_X(x) = \lambda e^{-\lambda x}$$

Laplace:

$$f_X(x) = \frac{1}{2b} e^{-|x-\mu|/b}$$

Gamma:

$$f_X(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}$$

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- Convert this into **clean LaTeX lecture notes**, or
- Make a **short exam revision version**, or
- Add **more solved problems** like your slide example.

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This scribe is based on your file **Lecture 8: Gaussian, Uniform, Exponential, and Gamma Random Variables**

CSE400 – Fundamentals of Probability in Computing

Lecture 8: Gaussian, Uniform, Exponential, Laplace, and Gamma Random Variables (Lecture Scribe)

Slide 1: Title and Context

Topic of the lecture:
Gaussian, Uniform, Exponential, Laplace, and Gamma Random Variables.

Purpose of this lecture:
To introduce the most important **continuous probability distributions** used in:

- Computing systems (noise, delays, errors)
- Communication systems (error probability, signal noise)
- Data modeling and statistics
- Performance and reliability analysis

We will learn:

- What a Gaussian random variable is and why it is important
- How to compute probabilities using special functions (Φ and Q)
- Other important continuous random variables and their uses
- How to solve probability questions step by step

Slide 2: Outline of the Lecture

The lecture is divided into these parts:

1. Gaussian Random Variable
- Definition and properties
- Standard forms: Error function and Q-function
- Relation between Φ-function and Q-function
- Example
2. Types of Continuous Random Variables
- Uniform Random Variable (with example)
- Exponential Random Variable (with example)
- Laplace Random Variable (with example)
- Gamma Random Variable
  - Graph and special cases
  - Example

3. Homework problem
4. Problem solving
5. In-class activity: Gaussian density estimation

Before starting, we recall:

A **continuous random variable (CRV)** can take **any real value in some interval**.
It is described using:

- PDF (Probability Density Function)
- CDF (Cumulative Distribution Function)

For a CRV X:

$$F_X(x) = P(X \leqslant x) = \int_{-\infty}^{x} f_X(t)\,dt$$

Slide 3: Moments, Skewness, and Kurtosis

Skewness (3rd standardized moment)

$$C_3 = \frac{E[(X-\mu)^3]}{\sigma^3}$$

Meaning in simple words:

- Skewness tells us **whether the distribution is symmetric or tilted**.
- If skewness > 0: the right tail is longer (distribution leans to the right).
- If skewness < 0: the left tail is longer (distribution leans to the left).
- If skewness = 0: the distribution is perfectly symmetric.

So, skewness is a **measure of symmetry of the PDF about the mean**.

Kurtosis (4th standardized moment)

$$C_4 = \frac{E[(X-\mu)^4]}{\sigma^4}$$

Meaning in simple words:

- Kurtosis tells us **how sharp or flat the peak** of the distribution is.
- Large kurtosis → very sharp peak near the mean (values are highly concentrated).

- Small kurtosis → flatter distribution (values are more spread out).

So, kurtosis is a **measure of peakedness** of the PDF.

Slide 4: Expected Value of a Function of a Random Variable

For a Continuous Random Variable (CRV)

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$$

Meaning:

- Multiply the function $g(x)$ with the PDF $f_X(x)$
- Integrate over all possible values of x
- This gives the **average (expected) value** of $g(X)$

For a Discrete Random Variable (DRV)

$$E[g(X)] = \sum_k g(x_k) P(X = x_k)$$

Linearity of Expectation

For constants a, b:

$$E[aX + b] = aE[X] + b$$

For a sum of terms:

$$E\left[\sum_{k=1}^{N} g_k(X)\right] = \sum_{k=1}^{N} E[g_k(X)]$$

Important logic:

- The expectation of a sum is the **sum of expectations**.
- This works **even if variables are dependent**.
- This is one of the most powerful tools in probability.

Slide 5: Start of Gaussian Random Variable

Now we start the **most important continuous distribution** in probability and statistics:
The **Gaussian (Normal) Random Variable**.

Slide 6–7: Definition and Properties of Gaussian Random Variable

Definition

A random variable X is called **Gaussian** if its PDF is:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

We write:

$$X \sim \mathrm{N}(\mu, \sigma^2)$$

Where:

- μ = mean (center of the distribution)
- σ^2 = variance (measure of spread)
- σ = standard deviation

Conceptual Meaning (from the figure in the slide)

- The PDF is **bell-shaped**.
- It is **symmetric** around $x = \mu$.
- Most values are near μ.
- As we move away from μ, the probability density decreases smoothly.
- The **total area under the curve = 1**.

The slide also shows:

- PDF: bell-shaped curve
- CDF: S-shaped curve (area accumulated from left to right)

Applications (as noted in the slide)

- Thermal noise in electronics

- Measurement errors
- Natural variations (heights, errors, random fluctuations)
- Signal processing and communication systems

Slide 8–10: Standard Forms – Error Function, Φ-function, Q-function

Error Function (erf)

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Why it is needed:

- Integrals of Gaussian functions **cannot be written in simple closed form**.
- So, special functions like **erf** are defined to express these integrals.

Complementary Error Function (erfc)

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt$$

Meaning:

- erfc gives the **remaining area** after erf.

Φ-function (CDF of Standard Normal)

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Meaning:

- $\Phi(x) = P(Z \leqslant x)$, where $Z \sim \mathrm{N}(0, 1)$.
- It gives the **area under the curve to the left of x**.

Q-function (Gaussian Tail Function)

$$Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Meaning:

- $Q(x) = P(Z > x)$.
- It gives the **area under the curve to the right of x** (right tail).

Important Relation (from slide note)

$$Q(x) = 1 - \Phi(x)$$

Because total area under the curve is 1.

Slide 11–17: Relation between Φ-function and Q-function (Derivation)

Let:

$$X \sim \mathrm{N}(\mu, \sigma^2)$$

We want to find the CDF:

$$F_X(x) = P(X \leqslant x)$$

Step 1: Write the definition of CDF

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right) dy$$

Step 2: Standardization (change of variable)

Let:

$$t = \frac{y-\mu}{\sigma}$$

Then:

- When $y = -\infty$, $t = -\infty$
- When $y = x$, $t = \frac{x-\mu}{\sigma}$

So the integral becomes:

$$F_X(x) = \int_{-\infty}^{(x-\mu)/\sigma} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Step 3: Recognize Φ-function

$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

So the CDF of any Gaussian can be written using the **standard normal CDF**.

Tail Probability

$$P(X > x) = \int_{(x-\mu)/\sigma}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt = Q\left(\frac{x-\mu}{\sigma}\right)$$

Key Identities (from slide)

$$Q(x) = 1 - \Phi(x)$$

$$F_X(x) = 1 - Q\left(\frac{x-\mu}{\sigma}\right)$$

Conceptual Explanation (from diagram slide)

- Φ(x) = area under the **left tail** up to x
- Q(x) = area under the **right tail** beyond x
- To evaluate CDF of Gaussian → use Φ-function
- To evaluate right-tail probability $P(X > x)$ → Q-function is more natural

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Slide 18–20: Worked Example

Given PDF:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Compare with standard Gaussian PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

We identify:

$$\mu = -3, \quad \sigma^2 = 4, \quad \sigma = 2$$

So:

$$X \sim \mathrm{N}(-3, 4)$$

---

1) Find $P(X \leqslant 0)$

$$P(X \leqslant 0) = F_X(0) = \Phi\left(\frac{0-(-3)}{2}\right) = \Phi\left(\frac{3}{2}\right)$$

Using relation:

$$= 1 - Q\left(\frac{3}{2}\right)$$

---

2) Find $P(X > 4)$

$$P(X > 4) = Q\left(\frac{4-(-3)}{2}\right) = Q\left(\frac{7}{2}\right)$$

General Rule for Intervals

$$P(a < X < b) = F_X(b) - F_X(a)$$

Types of Continuous Random Variables (from Outline)

Uniform Random Variable

PDF:

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leqslant x \leqslant b \\ 0, & \text{otherwise} \end{cases}$$

Mean:

$$E[X] = \frac{a+b}{2}$$

Variance:

$$\mathrm{Var}(X) = \frac{(b-a)^2}{12}$$

Application:

- Random number generation
- Simulation
- Modeling equal uncertainty

Exponential Random Variable

PDF:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geqslant 0 \\ 0, & x < 0 \end{cases}$$

Mean:

$$E[X] = \frac{1}{\lambda}$$

Variance:

$$\mathrm{Var}(X) = \frac{1}{\lambda^2}$$

Special property: **Memoryless**

Application:

- Waiting time problems
- Queueing systems
- Failure time modeling

Laplace Random Variable

PDF:

$$f_X(x) = \frac{1}{2b} e^{-|x-\mu|/b}$$

Properties:

- Sharper peak than Gaussian
- Heavier tails
- Used in noise modeling and privacy mechanisms

Gamma Random Variable

PDF:

$$f_X(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}, \quad x \geqslant 0$$

Mean:

$$E[X] = k\theta$$

Variance:

$$\mathrm{Var}(X) = k\theta^2$$

Special cases:

- Exponential distribution (k = 1)
- Erlang distribution (k is integer)

Application:

- Waiting time for multiple events
- Reliability and queueing models

Final Summary

- Gaussian distribution is the **most important continuous distribution**.
- It is defined by mean μ and variance σ^2.
- CDF is expressed using Φ-function.
- Right-tail probabilities are expressed using Q-function.
- Relation: $Q(x) = 1 - \Phi(x)$.
- Other important distributions: Uniform, Exponential, Laplace, Gamma.
- These are widely used in computing, communication, and data modeling.

Formula Sheet

Gaussian PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$$

Standardization:

$$Z = \frac{X - \mu}{\sigma}$$

CDF:

$$F_X(x) = \Phi\left(\frac{x - \mu}{\sigma}\right)$$

Q-function:

$$Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Relation:

$$Q(x) = 1 - \Phi(x)$$

Uniform:

$$f_X(x) = \frac{1}{b - a}$$

Exponential:

$$f_X(x) = \lambda e^{-\lambda x}$$

Laplace:

$$f_X(x) = \frac{1}{2b} e^{-|x-\mu|/b}$$

Gamma:

$$f_X(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}$$

If you want, I can now convert this into **clean LaTeX lecture notes**, or make a **short revision version**, or add **more solved examples** for practice.